$$\mathrm{f}(x) = \cos x + 2\sin x$$

(a) Express $\mathrm{f}(x)$ in the form $R\cos(x - \alpha)$, where $R$ and $\alpha$ are constants,

$R > 0$ and $0 < \alpha < \dfrac{\pi}{2}$

Give the exact value of $R$ and give the value of $\alpha$, in radians, to 3 decimal places.

**(3)**

$$\mathrm{g}(x) = 3 - 7\mathrm{f}(2x)$$

(b) Using the answer to part (a),

(i) write down the exact maximum value of $\mathrm{g}(x)$,

(ii) find the smallest positive value of $x$ for which this maximum value occurs, giving your answer to 2 decimal places.

**(3)**